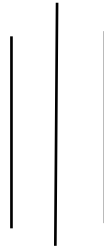




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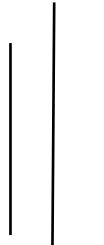
Kritipur, Kathmandu



An Internship Report On

“Development of Education Safari Website”

In partial fulfillment of requirement for the degree of
Bachelor of Information Management (BIM)



Submitted By:

Laxmi Regmi

Exam Roll No: 11001/19

TU Regd. No. : 7-2-240-45-2019

Saptagandaki Multiple Campus (SMC)

Bharatpur-10, Chitwan

September, 2024

DECLARATION

I hereby declare that this internship project entitled “Development of Education Safari Website” submitted to the Office of the Dean, Faculty of Management, Tribhuvan University, is a result of my own work carried out at Bitmap IT Solution Bharatpur, Chitwan for partial fulfillment of the requirements for the degree of Bachelor of Information Management (BIM). It has not been previously submitted to any other university or institution.

Laxmi Regmi

TU Roll No.: 11001/19

TU Regd. No.: 7-2-240-45-2019

ACKNOWLEDGEMENT

I would like to extend my heartfelt gratitude to Tribhuvan University and the Faculty of Management for providing me with the opportunity to participate in this internship program, which has been an invaluable part of my academic journey. My sincere thanks also go to Saptagandaki Multiple Campus for its continuous support throughout my internship.

I am profoundly grateful to Bitmap IT solution, which gave me the opportunity to work on the development of the Education Safari website. Being able to work in a real-world development environment has been a transformative experience, allowing me to put theoretical knowledge into practice. The exposure to modern technologies and the web development process will no doubt contribute immensely to my professional growth.

A special note of appreciation goes to my internship supervisor, Mr. Narayan Prasad Dahal sir, whose guidance, insights, and mentorship were pivotal in steering me through this project. Their valuable feedback and advice helped me navigate challenges and improve the quality of my work.

I would also like to express my deep gratitude to my project mentor, Mr. Sudip Parajuli sir, who patiently guided me through each phase of the development process. Their expertise and willingness to help at every step ensured that I could approach problems with confidence and find effective solutions.

Additionally, I would like to thank my family, friends, and colleagues for their unwavering support and encouragement throughout this internship. Their belief in my abilities kept me motivated, and their guidance and advice were indispensable in helping me complete this project successfully. This journey would not have been as enriching without their contributions.

Finally, I extend my gratitude to everyone involved directly or indirectly in making this internship a fruitful learning experience.

Laxmi Regmi

ABSTRACT

This report presents the development of *Education Safari*, a website designed to offer students a comprehensive platform for accessing educational resources. The website serves as an aggregator of academic information, including courses, learning materials, and other resources, aimed at helping students make informed decisions about their education. This report highlights the project's objectives, the process of designing and developing the website, and the methodologies employed throughout the internship.

During this internship, I was responsible for both the frontend and user experience design of the platform. The *Education Safari* website was developed using a range of modern web technologies, including HTML, CSS, JavaScript, Tailwind CSS and React JS ensuring that the platform was responsive, visually appealing, and accessible across multiple devices. Throughout the project, emphasis was placed on creating an intuitive user interface to enhance usability and navigation.

The project also involved overcoming real-world challenges related to website responsiveness, cross-browser compatibility, and user experience optimization. From designing the layout to implementing interactive features, this internship provided hands-on experience in web development. The tools used for design and development, including Figma for mockups and Git for version control, are discussed, along with the learning outcomes and skills acquired during the internship.

This experience allowed me to deepen my technical expertise, collaborate effectively within a professional environment, and deliver a functional product that aligns with modern web development standards.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BIM	Bachelor of Information Management
HTTP	Hypertext Transfer Protocol
IT	Internet Technology
JWT	JSON Web Tokens
KPI	Key Performance Indicator
NCCS	National College of Computer Studies
NTA	Nepal Telecommunications Authority
OOM	Object Oriented Model
PM	Project Manager
QA	Quality Assurance
SDEs	Software Development Engineers
SDLC	Software Development Lifecycle
TIBCO	The Information Bus Company
TU	Tribhuvan University
JS	JavaScript
BITS	Bitmap IT Solution

Chapter 1

INTRODUCTION

1.1 Background

Bachelor of Information Management (BIM) is a four-year course offered by TU that provides a unique blend of 60% information technology and 40% management courses which helps students to gain the technical expertise and various management concepts required in an organization. It is conducted by major Universities in Nepal in the semester system which is divided into eight semesters and 126 credit hours. This temporary job program has been planned for the halfway satisfaction of the level of BIM.

The objectives of the BIM are as follows:

- Production of professionals who have management skills as well as knowledge on Information Technology.
- Prepare IT professionals proficient in the use of computers and computational techniques in order to develop effective information systems to solve real life problems in the organizational environment.
- Production of IT professionals who are skilled at computers and computing techniques in order to create effective information systems which could solve the organization's problems.
- Develop ability in students in object-oriented software design methods and data management systems.

Internship is the situation of an understudy or student who works in an association here and there with or without pay so as to pick up work involvement or fulfill necessities for a capability. It is characterized as an understanding between an understudy and an association where an assistant fills in as a representative for the association for certain lime span. It is an impermanent position with an accentuation on hands on preparing instead of only business. It gives a stage to be engaged with a genuine working situation and create required vital aptitudes. The internship is one of the programs designed by the faculty of management, Tribhuvan University. It is a pre-professional work experience that provides students with the chance of gaining experience in a career field. This program helps us to know how the organization as a whole operates and also how communication takes place and activities are coordinated in the real world. Thus, the internship program can be regarded as a fruitful opportunity for the students of BIM to be familiar with the actual work environment of Nepal. An internship can give us a real insight into the world of work, allowing us to build on the theory we learned at university and helping us to gain practical skills that will

help strengthen our CV and make us more employable. Internships also offer the chance to test our skills in real-life situations, explore our career options and gain an insight into an organization or career path.

This report has been prepared based on the internship experience of 3 months in Bitmap IT Solution. It has provided real life experience and knowledge of how actually is the work place and work environment. The theoretical knowledge that I gained in college was applied in the actual field specifically in programming. This internship has not only sharpened my skills in programming, but also increased my self-confidence, self-motivation and self-development.

1.2 Focus of the Study

Internship is one of the most vital part in the student's life, who are about to start their career. It provides real life experience about the work, procedures and methodologies followed in the organization. This study focuses on the following areas:

Internship is one of the most vital part in the student's life, who are about to start their career. It provides real life experience about the work, procedures and methodologies followed in the organization. This study focuses on the following areas:

- Career opportunities

While working in an organization, students can get different opportunities to start their career. The knowledge they gain there can be very helpful in the future to work as an employee.

- Exposure

The students can get exposure while working in an organization. They get familiar with the working procedure and can be able to work under pressure. Also, they get in touch with various people who can help them in the future.

- Gain real life experience

One of the main purpose of internship program is to allow the students to gain real life experience. Students are restricted inside the classroom and do not get any platform to sharpen their skills. So internship provides a great platform where students can learn and showcase their knowledge and skills.

1.3 Statement of Problem

The primary challenge addressed by the Education Safari website was the lack of an organized, easily accessible platform for students to search and find relevant educational resources in one place.

a. **Fragmented Access to Educational Resources.**

Students often faced difficulties in locating quality educational materials, as information was scattered across various websites and platforms. This made it time-consuming and inefficient for them to find appropriate resources for their academic needs.

b. **Lack of a Centralized Repository.**

There was no centralized digital solution that aggregated courses, materials, and educational information tailored for students, which led to a lack of streamlined access. Students had to search multiple sources manually, often encountering outdated or irrelevant information.

c. **Limited User Experience in Existing Platforms.**

Existing platforms either lacked user-friendly interfaces or were not designed with the student's navigation experience in mind. Poor design made it harder for students to find what they were looking for quickly, leading to frustration and decreased engagement with the platform.

Through Education Safari, these problems were addressed by providing a comprehensive and responsive platform that centralized educational resources. The website offers a clean, intuitive interface, making it easier for students to access quality content efficiently and without the hassle of navigating multiple fragmented sources.

1.4 Objectives

1.4.1 Objectives of Internship

The most important objective of an internship is to make students familiar with the real working environment. By doing an internship, we get all of the personal and cultural perks while simultaneously reaping the professional benefits. The objectives of the internship program are as follows:

- To build interpersonal skills and confidence
- To implement the theoretical knowledge acquired from college in real working environment.
- To get career insight existing in the country.
- To receive an introduction to an organization within the Information systems industry and to gain a better understanding of its design and structure.
- To get familiar with the real working scenarios, and accordingly tackle them

1.4.2 Objectives of System

The *Education Safari* website was developed with the following objectives:

- To Centralize Educational Resources.

- To Enhance User Experience and Accessibility.
- To Ensure Responsiveness Across Devices.
- To Provide a Searchable and Organized Database.
- To Streamline the Process of Educational Resource Discovery.

1.5 Methodology

During the course of my internship, I collected information and requirements for the development of the Education Safari website through a combination of primary and secondary sources. The primary data was obtained through regular interactions with my project mentor and team members, along with feedback from potential users regarding their needs for educational resources. Day-to-day operations and team meetings provided valuable insights into how students search for and utilize educational materials online.

In addition, secondary data was collected by reviewing existing educational websites, analyzing competitor platforms, and exploring best practices for web design and development. Websites, research articles, user manuals, and online tutorials helped guide the technical aspects of the project, particularly in terms of using modern web technologies like HTML, CSS, JavaScript, Tailwind CSS and React JS.

The information collected during this process played a critical role in shaping the design and functionality of Education Safari, ensuring that it met the practical needs of students and provided a seamless user experience. Secondary data, such as design trends and user behavior studies, were easily accessible and contributed to making the website responsive, user-friendly, and visually engaging.

1.5.1 Organization Selection

The selection of an organization is a crucial factor for making the internship experience valuable and meaningful. It provides exposure to real-world environments and helps in applying theoretical knowledge in practical scenarios. For my internship, I chose Bitmap IT Solution, a company known for its expertise in web development and IT solutions. This decision aligned with my interest in website development and offered an opportunity to enhance my technical skills in a professional setting.

Bitmap IT Solution, with its established reputation in the industry, provided me with a platform to work on real-world projects and gain firsthand experience in web design and development. Throughout the internship, I was able to interact with experienced professionals, learn about modern web technologies, and contribute to the development of the Education Safari website, which focused on delivering a user-friendly platform for educational resources.

1.5.2 Placement and Duration

Organization's selection process placed me as an intern in the organization where I got the opportunity to learn the real organization process and software development process along with the importance of teamwork in achieving success. I was trained with the organization's work process that lead to an easy path for organization culture and process.

Table 1: Internship Schedule

Start Date	April 15, 2024
End Date	August 2, 2024
Days of Work	Sunday to Friday (6 Days)
Office Hours (Work from home)	9:00 AM to 6:00 PM (9 hours)
Position	Frontend Developer Intern
Internship Period	3 months

1.5.3 Work Procedure

For the initial week, I was introduced to the working environment of the company to make me feel comfortable. The second week was a mere orientation on designing concepts and design thinking. Third week went by with designing simple login templates, stepper bars and so like so on. I was given knowledge about the basics of front-end web development, persuasion and design psychology that are essential for carrying out the project more efficiently in the fourth week. I also learnt to operate various applications for carrying out my project smoothly which will not only help for the successful accomplishment of my project but will also provide guidance while doing some other projects. After 4 weeks of learning about the basics, I was given a design project that is being developed under React Js i.e. "Education Safari website". We were working from home due to lack of physical space inside the company's office. Yet, managed to have daily standups, meetings, project requirements collection and worked carried out in timely manner, just how we would physically in the office hours.

1.5.4 Tools Used

The Education Safari website was built using modern web development technologies to create a responsive, user-friendly, and scalable platform. The following tools were used in the development of the website:

- **React JS**

React.js is a powerful JavaScript library used to build the user interface (UI) of the Education Safari website. React's component-based architecture allowed for the development of reusable UI components, such as the navigation bar, search bar, and resource cards, which could be dynamically rendered across different pages of the website. This structure enhanced the website's performance by reducing unnecessary re-renders and optimizing user interactions. Additionally, React's virtual DOM efficiently handled updates to the UI, ensuring a fast and smooth user experience. The flexibility of React.js made it ideal for developing the interactive features of the website, including search functionality and filtering options.

- **HTML**

HTML (HyperText Markup Language) served as the backbone of the Education Safari website, providing the basic structure of the webpages. HTML was used to define the various elements on each page, including headers, buttons, forms, and links. In combination with React.js, HTML facilitated the creation of semantic, well-structured pages that are easy to navigate. Each section of the website, from the homepage to the resource listings, was crafted using HTML elements that were styled and made interactive using React and Tailwind CSS.

- **Tailwind CSS**

Tailwind CSS is a utility-first CSS framework that was used to style the Education Safari website. Unlike traditional CSS frameworks, Tailwind provides a set of utility classes that made it easy to apply specific styles directly within the HTML elements. This approach allowed for rapid development and ensured that the website maintained a consistent, responsive design across all devices. Tailwind CSS was instrumental in creating a clean and modern user interface, ensuring that the website was both visually appealing and easy to use. It also played a critical role in ensuring the responsiveness of the site, making sure that users could access the platform seamlessly across desktops, tablets, and smartphones.

1.6 Limitations Of Study

It was a great opportunity to be an intern in the IT Company. This report is prepared based on the observations, experience of the internship formal and informal interview with the staff and the secondary data available in internet. However, the internship had some limitations. The main limitations of internship are listed below:

- Due to lack of time constraint, study of each and every activity of the IT Company in depth was not possible.
- Due to security policy research was done in surface and not in depth.
- Due to privacy policy maintained in an organization there was unavailability of information.

Chapter 2

INTRODUCTION OF INDUSTRY

2.1 Introduction to Information Technology

Information technology is one of the growing sectors in Nepal which is just on the starting phase but has lots of potential for growth, investment and significant profits. Software companies are growing like mushrooms as users buy computers for home and business use and need something to do with them. Several industries are associated with information technology, including computer hardware, software, electronics, semiconductors, internet, telecom equipment, e-commerce and computer services. Both software development and the hardware involved in the IT industry include everything from computer systems to the design, implementation, study and development of IT and management systems. Owing to its easy accessibility and the wide range of IT products available, the demand for IT services has increased substantially over the years. The IT sector has emerged as a major global source of both growth and employment.

The government of Nepal has also identified IT as one of the five priority potential export service sectors in Nepal Trade Integration Strategy (NTIS) 2010. The IT industry has become an indispensable part of our economy as it helps in creation of more knowledge-based economy, human resource development, assist in effective management of other services and creates various other opportunities that aids in development of a nation. Using information technology, individuals and businesses have ability to view changes in global markets at real time far faster than they usually used to do.

2.2 Introduction to Software Industry

As we know that the world is moving from analog to digital, IT is playing an important role in people's lives. IT has become a part of life for humans. As the world is becoming narrower and narrower, the influence of IT is becoming more and more. There are 17 revolutions in all of the countries in the world, Due to the help of IT people, generating new techniques to improve their skills and talents. Nowadays, in all operations the involvement of information technology is vital.

Information Technology helps in building the nation's strength, and also provides a way to economic development. Today, information technology continues to advance and change the way of business works in unexpected new ways. IT can be defined as a contemporary term that describes the combination of computer technologies with telecommunication technologies. Its vast power to affect change has put it in

the forefront of all other industries. Today, IT is the fastest growing economic activity in the world and IT industry as such is the biggest industry.

IT has helped computerize the business process thus streamlining businesses to make them extremely cost-effective money-making machines. Thus, in turn increases productivity which ultimately gives rise to profit that means better pay and less strenuous working conditions. IT has made it possible for businesses to be open 24 x7 for business all over the globe.

The IT industry can serve as a medium of e-governance, as it assures easy accessibility to information. The use of information technology in the service sector improves operational efficiency and adds to transparency. It also serves as a medium of skill formation. In this new era, IT stands as the central force in shaping organizations, societies and nations based on the presumption that it is the key to achieve progress today. At last, Information technologies are the product of developed countries and to make that technology suitable for developing countries like Nepal. There should be an effort to build a capacity to recognize the importance of implementing IT according to the development needs and requirements. So, IT is the most vital component to build up the nation and helps economic growth.

2.3 Objectives of Information Technology in Business

- To ensure the availability of and access to information that enables customers to make timely, informed decisions by strengthening data and knowledge management approaches, ("Department IT Strategic Goals", 2010).
- To explore new horizons of challenges and opportunities that can boost the technological, economic and educational scenario of the country.
- To manage the cost efficiencies
- To provide a robust and secure IT infrastructure that supports on-demand access to information
- Globalization.
- Promoting the growth of e-commerce and software export.
- To improve end-to-end transparency and accountability between the customer and service provider.
- To increase percentage of employees who enroll in IT training opportunities

2.4 History of IT in Nepal

In this revolution of IT, we have witnessed many changes from machine era to automated era. Coming to this long we are now totally living in a world of technology. A brief history of IT in Nepal is shown below:

- Use of computer for the first time in history of NEPAL in 1971 census by NCC.
- Another 2nd generation computer was used in 1981 census named ICL 2950/10.
- Telephone Service exchange was established by NTC in 1960 in Kathmandu Valley.
- RONA E-Mail Service.
- Mercantile Communications in 1995 started Internet Service for the 1st time.
- In 1998, NTA, a telecommunications regulatory body, was formed as per the Telecommunications Act 1997, (Dhungel, 2019).
- Involvement of ISPs.
- IT policy 2000 - place Nepal on global IT map, to make IT accessible to general public, creation of knowledge-based society/industries, (Bhurtyal, 2010)
- Telecommunication Policy 2004 - to create favorable environment to make the telecommunication service reliable and accessible
- eGovernment Master Plan 2005
- Electronic Transaction act 2008 - ADB supported ICT Development Project

Due to the success of NCC's, there are more than 300 legal private computer training institutes. I.T Ministry established for advising and formulating plans and policies by NIC instead of NCC. NTC provide cellular and WLL network to private parties UTL, NCELL, Hello Nepal. NTA provide license to various ISPs like- World link Communications Private Limited Broad link Network and Communications Private Limited Hotlink Nepal Private Limited. Traditional methods of business and education are now replaced by advance I.T innovations, (Soft, 2019). Rapid and efficient development of Telecommunication and transmission media. Now there are more than 4 million users of the internet. Although there are a lot of encouraging signs for Nepal in the field of IT, the current quality of IT services and education still leaves a lot to be desired. A considerable amount of work needs to be done, and the initiative should be taken early if we are to level and keep up with the world in the field of technologies. The 21st century demands globalization.

2.5 Opportunities in Nepali IT Sector

The Nepali IT sector is experiencing rapid growth, offering a range of opportunities for professionals and businesses alike. As the demand for digital transformation increases across various industries, the IT sector in Nepal is becoming a critical driver of innovation, economic development, and job creation. With advancements in web development, mobile applications, cloud computing, and software solutions, the IT industry is providing new pathways for young professionals to build their careers.

One of the major opportunities lies in the growing need for digital platforms, such as e-commerce, education, and healthcare services. The rise of online education platforms like Education Safari highlights the growing demand for digital solutions in the education sector. This trend has created numerous opportunities for IT professionals to develop user-friendly, scalable, and responsive websites and applications that cater to a tech-savvy audience.

Furthermore, the government's push toward digitization, coupled with increasing internet penetration, has expanded the scope of IT services. Opportunities also exist in areas such as cybersecurity, data analytics, artificial intelligence (AI), and mobile development, which are becoming increasingly important as businesses move towards a digital-first approach.

Nepal's growing pool of skilled IT graduates, coupled with the availability of affordable resources, positions the country as an emerging hub for outsourcing IT services to international markets. This not only boosts the local economy but also opens up opportunities for IT professionals to work on global projects from Nepal.

2.6 Challenges in Nepali IT Sector

Despite the rapid growth and emerging opportunities in Nepal's IT sector, several challenges continue to hinder its full potential. These challenges range from infrastructural limitations to skill gaps and regulatory issues, all of which need to be addressed to sustain long-term growth and competitiveness.

One of the primary challenges is the lack of reliable infrastructure, particularly in terms of stable electricity supply and high-speed internet connectivity. Although internet penetration has been increasing, many rural areas still suffer from unreliable connections, which limits the reach of digital platforms and online services, such as education platforms like Education Safari. This also affects the ability of businesses and professionals to operate efficiently in an increasingly globalized and digital economy.

Another significant challenge is the shortage of skilled professionals with advanced knowledge in fields such as cybersecurity, artificial intelligence, and data science. While there are many talented IT graduates in Nepal, the fast-evolving nature of technology often outpaces the available training and education. This skill gap creates difficulties for companies in finding the right talent, leading to a dependency on foreign expertise or outsourcing for complex projects.

Additionally, there are regulatory challenges, including inadequate policy frameworks and a lack of government incentives for IT startups and innovation. Bureaucratic red tape, tax issues, and unclear guidelines around intellectual property protection can make it difficult for IT companies to operate smoothly and scale their businesses.

Moreover, the IT sector in Nepal faces stiff competition from neighboring countries such as India, which is already established as a global IT hub. This competitive pressure, combined with lower investment in research and development (R&D) in Nepal, makes it challenging for local companies to keep up with international standards and technological advancements.

Addressing these challenges will require coordinated efforts from both the government and private sectors to improve infrastructure, update education systems, provide incentives for innovation, and develop a more conducive environment for IT professionals and businesses to thrive.

Chapter 3

INTRODUCTION OF ORGANIZATION

3.1 Introduction of Organization

BITS is a creatively driven integrated Business Consulting, Advertising agency and IT company located in Narayangarh, Chitwan, Nepal that creates high-impact business solutions for both traditional and digital media to its clients. Bitmap pursues the dexterous technique to isolate the errands into littler dashes of one week. They take a shot at the premise of customary criticism with the client, executing new advances for a customer to meet their prerequisites effectively and bringing people together as efficient, highly collaborative teams to continuously deliver value to users.

They believe in providing cost effective and reliable services to their clients with their highly skilled team members. Their expert team is always ready to serve you with professional custom website design, web development, graphics design, digital marketing, mobile app development, e-commerce website design and development, community websites design, website redesign etc. and also the maintenance service for your website like domain and hosting. Their experts are well skilled and can build stylish as well as elegant and easy navigated websites for you. Over the years of experience, Bitmap IT Solution has achieved a prominent position of an expert software company possessing some of the best analytical brains. Their transparent, efficient and flexible world class software development process zero downs risks of project failures and creates powerful software solutions that meet present as well as future demands.

3.2 Vision and Scope

Customer loyalty through customer satisfaction, delivery in time, quality assurance and management, innovation to improve efficiency, live support and to be recognized as the best company when it comes to delivering value to customers.

3.3 Services Offered

Primary services offered by BITS are:

- **UI/UX and Graphics Designing**

BITS has been offering creative UI/UX and graphic design solutions to a variety of business needs. It offers a complete array of graphic services that combine strategy. Creative thinking and technical expertise in making designs stand out and deliver. It also provides an absolute solution to the businesses or individuals, to post simple information about themselves or about their company onto the web page.

They have been providing a complete dynamic web solution as per the client's expectations. It designs and develops dynamic web sites as per the specific requirements of clients which are flexible, user friendly, with effectiveness and a great amount of interactivity. The prototyping service really makes it easier for both clients and developers.

- **Software Development**

BITS has been providing custom software using various programming languages like C, C#, java, react and .NET. They deliver rapid application development processes and thoroughly tested and completely documented systems that can be deployed on any major IT platform. Whether you want a single or multi-user desktop application, or an enterprise-wide server-based solution, BITS have the knowledge and experience to get the job done.

- **Digital Marketing.**

When correctly executed, online marketing can be one of the most cost-effective and efficient ways of reaching your target audience. The online marketing services include search engine optimization (SEO), pay-per-click (PPC) advertising, email marketing, social media marketing, Google Analytics, Google Website Optimizer and training.

- **Search Engine Optimization**

Search engine optimization is a methodology of strategies, techniques and tactics used to increase the number of visitors to a website by obtaining a high-ranking placement in the search results page of a search engine including Google, Bing, Yahoo and other search engines. BITS implements ethical and effective search engine optimization strategies for websites so that it can enjoy top page ranking in Google and other major search engines naturally building on online presence for the site through social media as well.

- **Web Hosting**

Web hosting is the activity or business of providing storage spaces and access for websites. BITS servers are Linux-based, high-end Web servers designed from the ground up with small business requirements in mind. The combination of the Linux server platform and the state-of-the-art data center translates into the most stable environment available for your small business website, email accounts and databases.

Other services offered by BITS are:

- **E-Commerce Application Development.**

- Integrated Payment System.
- Content Management Solutions.
- Web Portal Development.
- Flash Developments.
- Graphics Designing.
- Training Classes.
- Domain and Hosting.
- Internships.
- 2D Animation.
- Mobile App Development.

3.4 Organization Structure

Organization structure is a system that is used to define a hierarchy within an organization. It helps to run the organization in a smooth manner. Therefore, BITS has a systematic structure consisting of CEO, Technical Director, Developer, Senior Programmer and Junior Programmer who handle different field of the organization.

BITS has adopted a structured hierarchy to allocate roles effectively, dividing the workforce into different levels based on their expertise and responsibilities. The main positions in BITS are as follows:

- **Chief Executive Officer (CEO):** The CEO is the highest-ranking individual in the organization and is responsible for the overall management and direction of the company.
- **Technical Director:** Reporting directly to the CEO, the Technical Director oversees all technical aspects of the organization.
- **Developer:** Developers are responsible for the actual building and coding of software, websites, or applications.
- **Senior Programmer:** Senior Programmers are more experienced developers who lead coding and development efforts.
- **Junior Programmer:** Junior Programmers assist Senior Programmers in developing software and applications. Their role involves writing basic code, debugging, testing, and learning from more experienced colleagues.

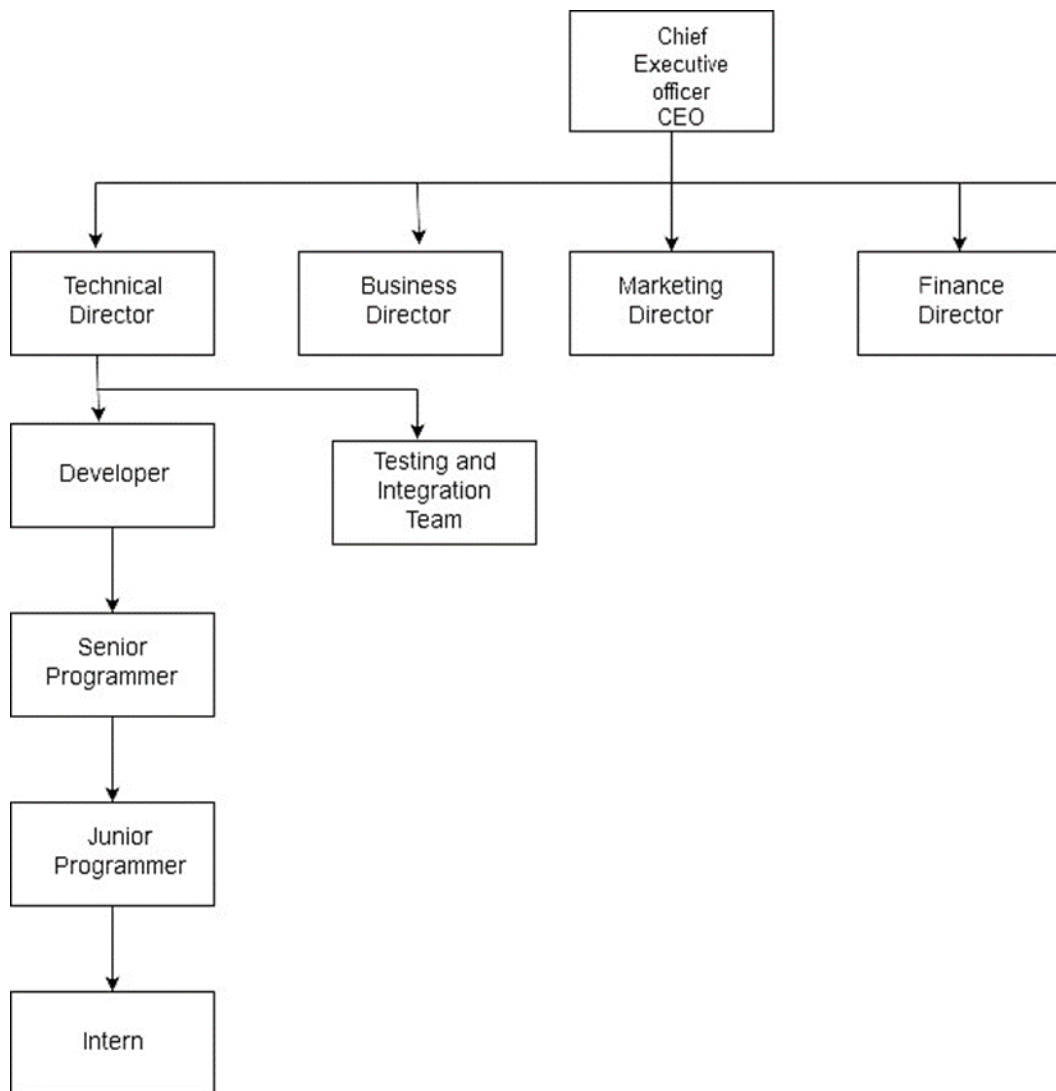


Figure 1: organization Structure

Chapter 4

SYSTEM ANALYSIS, DESIGN AND DEVELOPMENT

4.1 Introduction to the System.

Education Safari is a web-based platform designed to aggregate and provide easy access to a wide range of educational resources. The primary goal of this system is to simplify the process of finding and accessing academic materials, courses, and other educational tools, creating a one-stop platform for students and learners. The system allows users to browse, search, and filter educational resources based on their needs, ensuring that they can quickly locate relevant information in an organized and efficient manner.

The development of this platform focuses on enhancing the user experience by providing an intuitive, responsive, and visually appealing interface. The website was designed with the aim of reducing the time and effort students spend navigating multiple sources to find reliable academic resources. Education Safari streamlines this process by centralizing various types of educational content in one platform.

The system not only offers a user-friendly environment for browsing and searching resources but also ensures that all data and resources are well-organized and easily accessible. The platform's responsive design ensures compatibility across various devices, allowing students to access the resources they need anytime and anywhere.

Education Safari provides an organized database of educational resources and incorporates features that improve user engagement, such as interactive search options and a clean interface. The system is built using modern web technologies like HTML, CSS, JavaScript, Tailwind CSS and React JS to ensure optimal performance, scalability, and ease of maintenance.

4.2 Objectives of the System

- To Centralize Educational Resources.
- To Improve Accessibility.
- Enhance User Experience.
- To Facilitate Resource Discovery.
- To Promote Digital Learning.
- To Ensure a Scalable and Secure Platform.

4.3 Functions of the System

- Resource Aggregation.

- Search and Filtering.
- Responsive Design.
- User-Friendly Interface.
- Content Categorization.
- Data Security and Privacy.

4.4 Requirements of the System

4.4.1 Functional Requirements

These are the essential functions that the Education Safari system must provide, how it should react to specific user actions, and how it should perform in different situations. The Education Safari website should be able to perform the following activities:

- Users must be able to search for courses, materials, and resources and apply filters based on topics, subjects, and educational levels.
- The system should allow users to easily navigate through content organized by categories, including subjects, courses, and levels.
- The website must automatically adapt to different screen sizes and devices (desktop, tablet, and mobile) for an optimal user experience.
- Each user must have the ability to create an account, login/logout, and manage their profile, including updating personal details and saved resources.
- The system should provide resource recommendations based on users' search history and preferences to improve content discovery.
- Admin users must be able to upload, edit, update, or remove educational resources to ensure the platform remains current and relevant.
- Users and administrators should interact with the platform through secure authentication protocols, ensuring data privacy and protection.
- The platform should allow users to bookmark or save resources for future reference, with an option to easily access them from their profile.

4.4.2 Non-Functional Requirements

Non-functional requirements are requirements that are not directly concerned with the specific delivery by the system to its users. They may relate to emergent system properties such as reliability, response time and store occupancy. NFR may come from required characteristics of the software (product requirements), or from external sources. They are:

- Performance:
 - Response Time: The website should load within 2-3 seconds for optimal user experience.
 - Scalability: The system should handle up to 10,000 simultaneous users without performance degradation.
- Availability:
 - Uptime: The website should have an uptime of 99.9% to ensure continuous availability.
 - Maintenance: Scheduled maintenance should be communicated in advance and occur during off-peak hours.
- Usability:
 - Accessibility: The website should comply with WCAG 2.1 Level AA accessibility standards to ensure it is usable by people with disabilities.
 - User Interface: The UI should be intuitive and user-friendly, with a clear navigation structure and responsive design.
- Security:
 - Authentication: The system should use secure authentication mechanisms such as OAuth or JWT for user and admin access.
 - Data Encryption: All sensitive data should be encrypted both in transit (using SSL/TLS) and at rest.
 - Data Privacy: The system should adhere to GDPR or relevant data protection regulations to safeguard user information.
- Compatibility:
 - Cross-Browser Support: The website should function correctly on major web browsers (e.g., Chrome, Firefox, Safari, Edge).
 - Device Compatibility: The website should be fully functional on various devices including desktops, tablets, and smartphones.
- Maintainability:
 - Code Quality: The codebase should follow the best practices for maintainability, including proper documentation and code comments.
 - Modularity: The system architecture should support modular updates and feature additions with minimal impact on existing functionality.

4.5 Feasibility Analysis

The main objective of feasibility study is to test the technical, operational, economic and schedule feasibility. All systems are feasible only if they are given unlimited resources and infinite time. It helps to determine the benefits of the proposed system in society and organization. It also determines if the system can be built successfully with cost, time and effort.

4.5.1 Economic Feasibility

The Education Safari website is economically feasible as it utilizes cost-effective web technologies such as HTML, CSS, JavaScript, Tailwind CSS and React Js all of which are open-source and free to use. The platform's development primarily requires resources like a stable internet connection, basic computing hardware, and web hosting services, all of which are readily accessible at minimal cost. Additionally, the platform is designed to be scalable, meaning future updates or expansions can be made without incurring significant additional expenses.

4.5.2 Operational Feasibility

The Education Safari website is highly feasible from an operational perspective. The platform is designed with a simple, intuitive user interface that ensures ease of navigation for students, teachers, and administrators. Although some users may initially be reluctant to shift to a new platform for accessing educational resources, the ease of use and organization of the platform will help them adapt quickly. Moreover, the system's cross-device compatibility, which includes seamless functionality on desktops, tablets, and smartphones, ensures that users can access resources anytime, anywhere. Overall, the system's operational features align well with its objectives, making it highly feasible for regular use in educational settings.

4.5.3 Technical Feasibility

The technical feasibility of the Education Safari website is strong, as it leverages widely available web technologies such as HTML, CSS, JavaScript, and Tailwind CSS. These technologies are well-supported across all modern browsers and devices, ensuring smooth performance. The platform's architecture is lightweight, requiring only a stable internet connection to access. The necessary technical resources, such as development tools, databases, and hosting infrastructure, are readily available, making the system feasible from a technical standpoint. The platform is designed to be scalable, which allows for future expansion, including the addition of new features or resource categories, without significant technical hurdles.

4.5.4 Schedule Feasibility

The development schedule for Education Safari was carefully planned to ensure timely completion of each project phase. The website's features, such as resource aggregation, search and filter options, and a responsive interface, were developed and tested within the scheduled timeframe. With future enhancements or updates in mind, the system's modular structure ensures that new features or updates can be implemented efficiently without disrupting existing functionality. This makes the project highly feasible from a scheduling perspective, allowing for both present and future development needs to be met within reasonable time constraints, ultimately saving both time and cost.

4.6 System Design and Modeling

4.6.1 Process Modeling

Process modeling (UI flow) involves mapping out how users interact with the system, outlining the steps they take to navigate through the website. In the case of Education Safari, process modeling would describe how a user moves from one section of the website to another, such as from the homepage to searching for resources, filtering results, and viewing resource details. It helps ensure that the user journey is intuitive, smooth, and efficient. UI flow diagrams or site maps can be used to visualize this process.

- **UI Flow Diagram:** A diagram that shows the flow of user interactions between different pages, such as from the homepage to search results, and then to a resource details page. This can be a simple flowchart that visually represents how users navigate through the website.

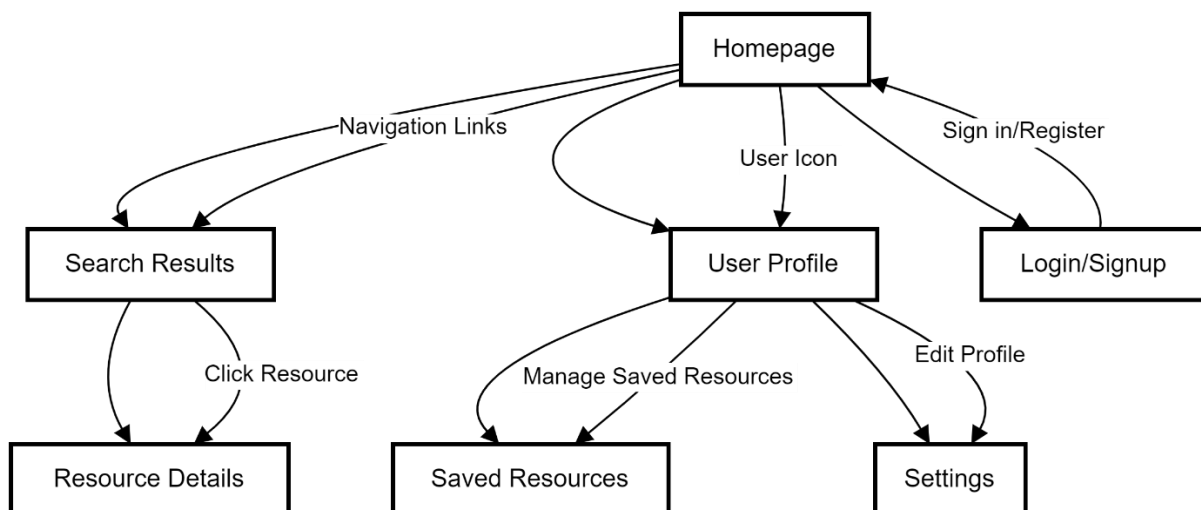


Figure 2: UI Flow Diagram

4.6.2 User Interface Design

User Interface (UI) design focuses on creating the visual elements that users interact with on the website. It includes the layout, typography, color schemes, buttons, and overall aesthetics of the platform. In Education Safari, UI design ensures that the website is visually appealing, easy to navigate, and responsive across different devices (desktop, tablet, mobile). Tools like React.js and Tailwind CSS were used to build and style the interface, ensuring a consistent and user-friendly experience.

- Wireframes of the key pages like the homepage, search results, and resource details.

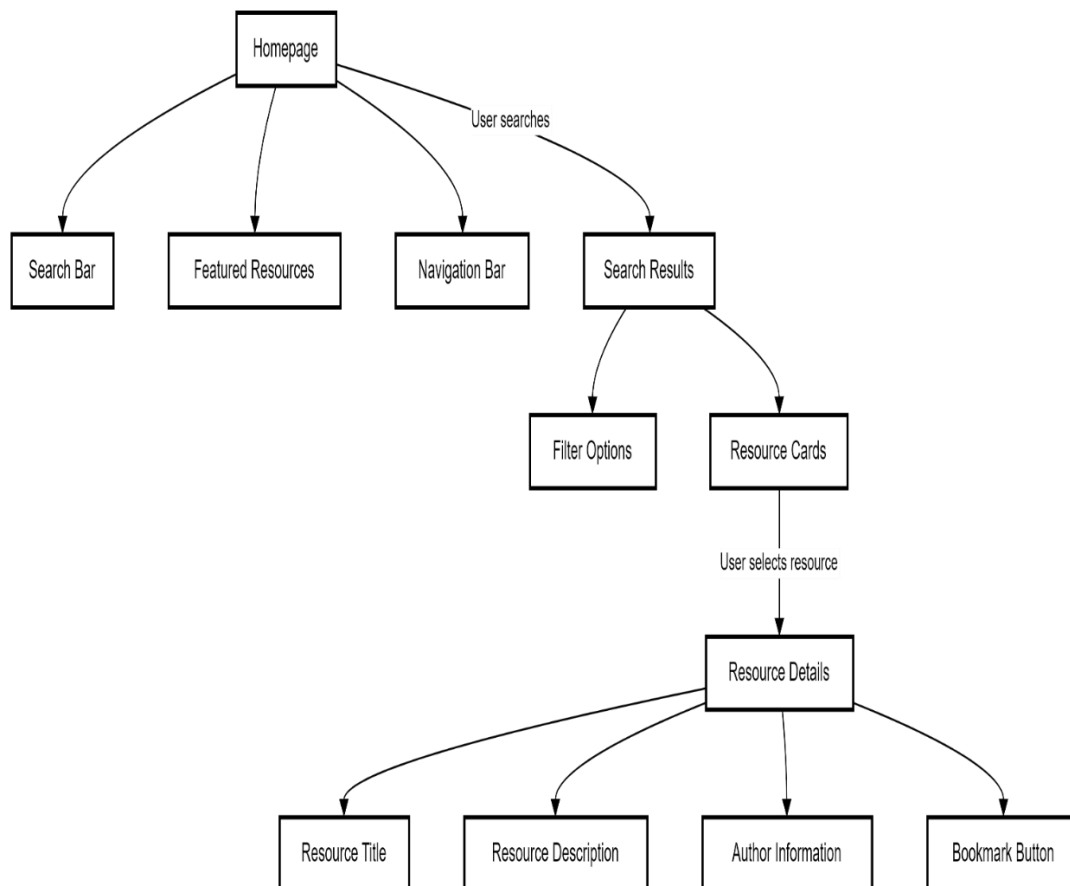


Figure 3: Wireframe For System

4.6.3 Component-Based Architecture (React.js)

React.js follows a component-based architecture, meaning that the user interface is broken down into small, reusable components. In the case of Education Safari, each part of the website—such as the navigation bar, search bar, resource listings, and footer—is developed as a separate component. This approach makes the development more modular, allowing each component to be independently maintained and reused throughout the application. For instance, the App component serves as the main container, while sub-

components like NavBar, Search Bar, ResourceList, and Footer handle specific parts of the UI. This architecture enhances maintainability and scalability since components can be reused, updated, or removed without affecting the rest of the system.

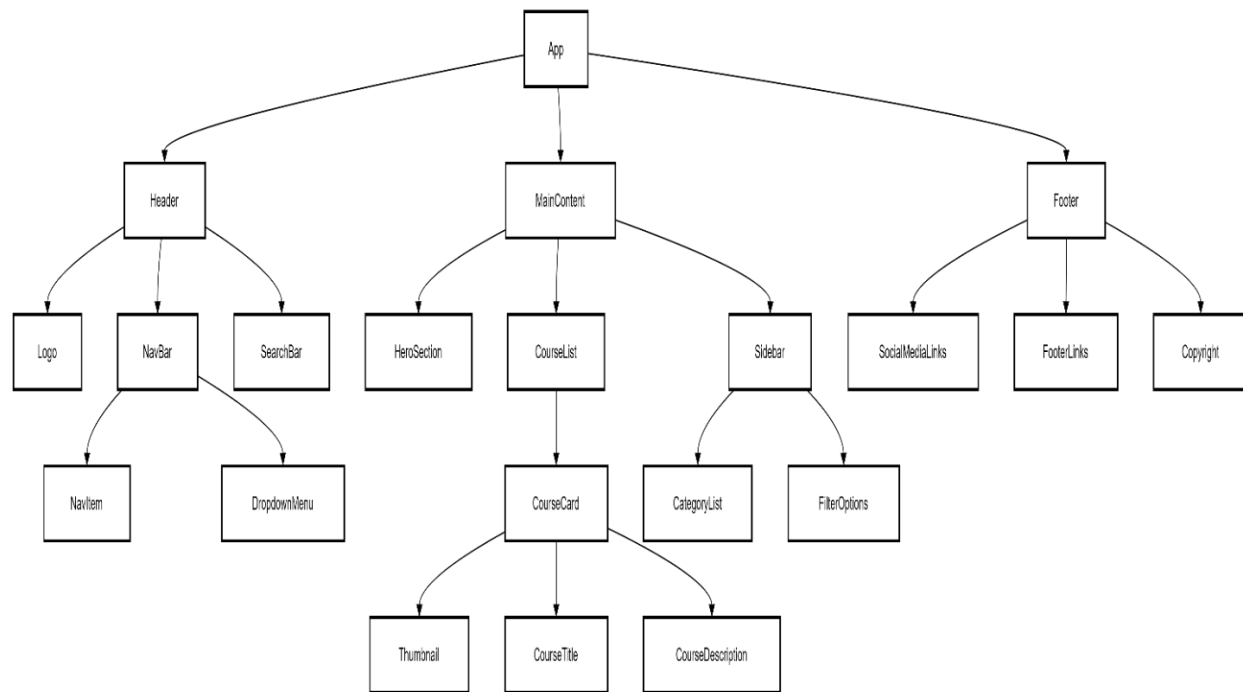


Figure 4: Component Hierarchy Diagram

4.6.4 State Management (React.js)

In React, state management refers to how data is managed and passed between components. For Education Safari, managing state is critical, especially for dynamic data like search results, user inputs, and resource filtering. React's built-in `useState` and `useEffect` hooks were used to handle component state. For example, when a user enters a search query, the input is captured and stored in the state. This state is then used to fetch data (such as search results) and display it in the UI. If external state management was necessary for handling complex states, Redux or Context API might have been used to manage global state across the entire application.

4.6.5 Routing and Navigation (React Router)

In Education Safari, routing and navigation were handled using React Router, a library that allows for efficient client-side routing. This means that different pages—such as the homepage, search results page, and resource details page—are dynamically rendered without requiring a full page reload. React Router manages routes like `"/"`, `"/search"`, and `"/resource/"`, ensuring users can easily navigate between sections of

the site. Each route is linked to a specific component that renders the appropriate UI for that page. For instance, the Home component is rendered when users visit the root route ("/"), while the Resource Details component is rendered when users navigate to "/resource/".

4.6.6 Responsive Web Design (Tailwind CSS)

Responsive web design ensures that the website looks good and functions well on devices with different screen sizes (desktop, tablet, mobile). In Education Safari, Tailwind CSS was used to implement responsive design by applying utility classes such as md:flex, lg:grid, and sm:block, which adjust the layout based on screen size. By using Tailwind's responsive utility classes, the layout automatically adjusts, ensuring the site remains user-friendly on all devices. For example, the navigation menu might collapse into a hamburger menu on mobile screens, and resource cards might stack vertically rather than displaying in a grid.

4.6.6.1 Large Screen Device

For large screen devices such as desktops, Tailwind uses the lg: prefix to target screen sizes 1024px and above. This allows designers to implement grid layouts or flexbox systems that take full advantage of the larger display area, making navigation menus and other content fully visible and horizontally aligned.

4.6.6.2 Medium Screen Device

For medium screen devices like tablets, the md: prefix (768px to 1023px) is used. Here, layouts may remain flexible, but content often shifts to accommodate the smaller space. For example, a md:flex class can allow for elements to rearrange and resize fluidly while ensuring that navigation menus are still fully visible, albeit in a slightly more compact format.

4.6.6.3 Small Screen Device

For small screen devices, such as mobile phones, the sm: prefix (640px to 767px) helps to simplify the layout. Elements might stack vertically using classes like semiblock, and navigation menus typically collapse into a hamburger format for easier access. Resource cards and content sections are often displayed in a single-column view, optimizing readability and usability on smaller screens. By using these responsive utility classes, Tailwind CSS ensures that a website's layout adapts smoothly to any screen size, enhancing user experience across all devices.

4.7 System Development Lifecycle (SDLC)

SDLC is a framework that development teams use to produce high-quality software in a systematic and cost-effective way. The SDLC methodology is used by both large and small software organizations. These teams follow development models ranging from agile to lean to waterfall and others. The software development lifecycle gives organizations a methodical, step-by-step approach to developing successful

software. From gathering the initial requirements for a new product, through maintaining a mature product on the market, SDLC is important, ("SDLC Overview", n.d.).

Objectives of SDLC are as follows:

- To establish appropriate levels of management authority to provide timely direction, coordination, control, review, and approval of the system development project
- Documenting requirements and maintaining trace ability of those requirements throughout the development and implementation process
- Ensuring that projects are developed within the current and planned information technology infrastructure (Software Development Life Cycle, 2016).

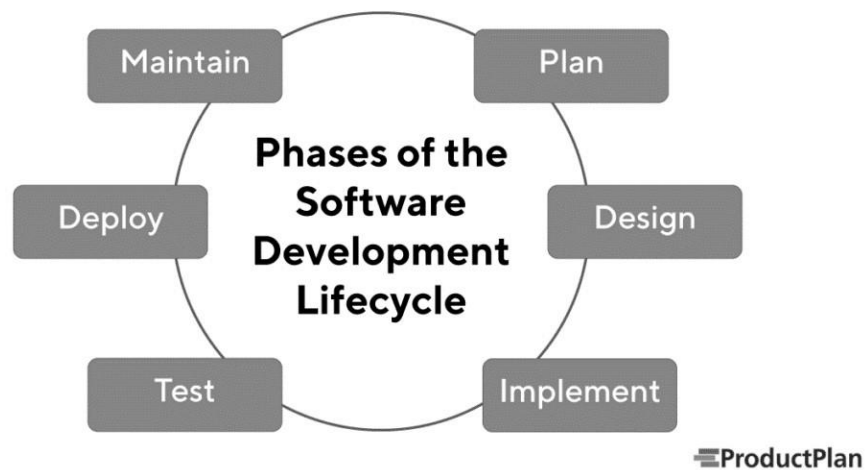


Figure 5: System Development Life Cycle

4.8 Prototyping Model

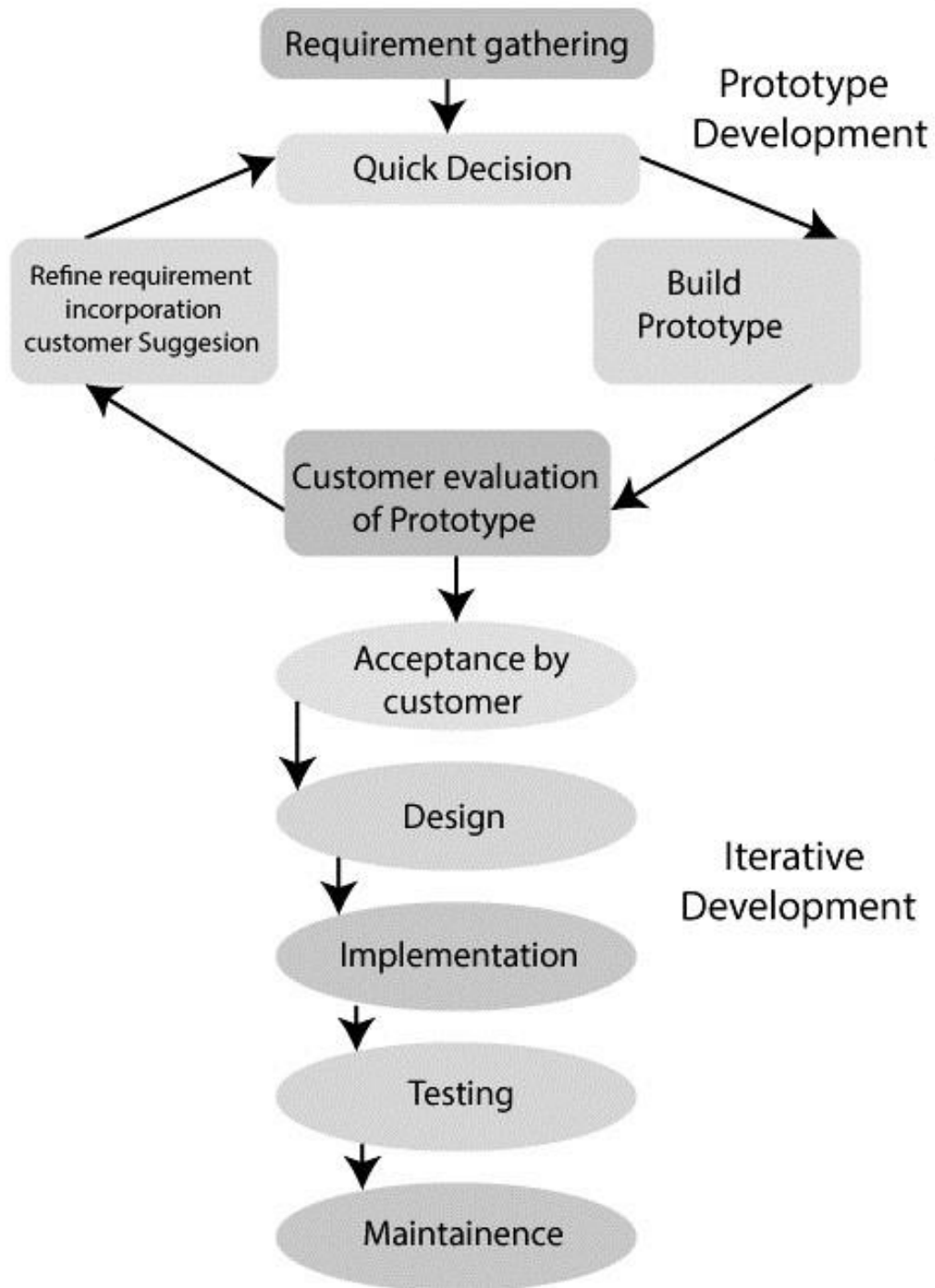


Figure 6: Prototyping Model

4.8.1 Requirement Analysis

Also called planning and analysis, the first stage of new software development will be to gather all relevant information from stakeholders and analyze this information to determine what will be feasible. This includes compiling requirements, studying user personas, and agreeing on the product's purpose. During this phase, the team will also discuss the opportunities and risks of pursuing the project. Here in Education Safari website too, the requirements were gathered, as discussed in earlier sections. They were analyzed before starting any other activities. Even requirement feasibility analysis was done, and the risky parts or unachievable parts were cut off in the first stage itself.

4.8.2 Design

After gathering and confirming the project requirements, the next phase was designing the Education Safari website, focusing on both the user interface (UI) and user experience (UX). The design phase began by asking, "How will we structure and build this platform?" This phase included creating wireframes and mockups to visualize the system layout and ensure that all functionalities would be easy to access and user-friendly.

For the initial design phase, Figma was used as the primary tool for creating both low-fidelity and high-fidelity prototypes. Figma allowed for collaborative designing, where different design options for the platform's interface were explored. Starting with basic wireframes, I mapped out the layout of key sections like the homepage, search interface, resource categories, and user profile pages. The wireframes focused on organizing content in a clean and logical way to provide easy navigation and clear access to resources.

Once the wireframes were approved, they were turned into high-fidelity prototypes, where details such as color schemes, typography, and UI elements were finalized. The platform was designed with a focus on readability and simplicity, using clean fonts and a color scheme that is visually appealing without overwhelming users. Special attention was given to ensuring that the design was responsive, meaning it would work seamlessly across various devices, including desktops, tablets, and mobile phones.

In addition to the visual aspects, user flows were designed to ensure that navigating the website would be intuitive. For instance, the search bar and filtering options were prominently positioned to make it easy for users to find the resources they needed. Mockups of different interactions, such as saving resources to a profile or browsing through categories, were created to ensure that all possible user actions were covered.

The prototypes underwent multiple rounds of testing with potential users to gather feedback on usability and navigation. Adjustments were made based on the feedback, such as improving the placement of certain buttons, enhancing the visibility of search options, and refining the mobile version of the site. Only once

the design was thoroughly tested and approved, the development process began, minimizing risks of changes during the later stages of the project.

4.8.3 Development

The development of the Education Safari website involved the integration of modern web technologies to create a robust, scalable, and user-friendly platform. The frontend of the website was built using React.js, a JavaScript library known for its efficiency in building interactive user interfaces, while Node.js was used on the backend for handling server-side operations and API development. This combination of technologies ensured that the system would be responsive, scalable, and capable of handling multiple user requests efficiently.

The primary objective during development was to build a seamless and responsive user interface for the educational platform. React.js was chosen for its component-based architecture, which allowed us to create reusable UI components such as the navigation bar, search bar, resource cards, and user profile management. React's virtual DOM ensured fast rendering of these components, which significantly enhanced the website's performance, especially for users navigating between different resource pages.

On the backend, Node.js was used along with the Express.js framework to handle server-side logic and manage API endpoints. These APIs were responsible for handling requests like fetching educational resources, managing user profiles, and handling searches. The REST API endpoints were built to interact with the frontend by accepting requests and sending the appropriate data back. For example, when a user searches for a specific resource, the frontend (React.js) sends a request to the backend (Node.js), which processes the query, retrieves the data from the database, and returns it in a structured format for display on the frontend.

For the database, MongoDB was used as the NoSQL database management system. MongoDB's flexibility allowed us to store various types of data, including user profiles, saved resources, and educational content. The structure of the database was designed to avoid redundancy while ensuring quick access to frequently queried data, such as resources categorized by subjects or user preferences. Collections were created to manage user information, resources, and category-based filtering.

One of the key challenges during development was ensuring that the website was responsive and could handle large amounts of content without performance degradation. To solve this, React's lazy loading feature was implemented to load only the necessary components as needed, which reduced initial load times and improved overall performance. Additionally, CSS-in-JS libraries like Tailwind CSS were used to manage styling within React, ensuring consistent styling across different components and enhancing the maintainability of the codebase.

In terms of security, JWT (JSON Web Tokens) was implemented for user authentication and authorization. This ensured that only authenticated users could access personalized features like saving resources and managing profiles. Users could log in, and upon successful authentication, they received a token that would grant them access to protected routes on the website. This token-based approach also allowed for secure communication between the frontend and backend.

During the development process, Git was used for version control, enabling efficient collaboration and tracking of changes. All changes were pushed to a shared repository, allowing the team to work concurrently on different features and resolve issues quickly. Once the development was complete, the website was deployed using Heroku for hosting the backend and Netlify for the frontend. This deployment strategy ensured smooth hosting and fast performance, with automatic updates whenever changes were pushed to the repository.

Overall, the development of Education Safari involved integrating powerful tools and technologies to ensure the platform met user expectations, provided fast and secure access to educational resources, and could be scaled for future growth.

4.8.4 Testing

Testing is an essential phase of the development process that ensures the functionality of the Education Safari website aligns with the specified requirements. The goal of testing is to identify and resolve any bugs or inconsistencies in the system early, preventing potential issues during live usage. Early detection of bugs saves time and costs associated with fixing problems later in the development cycle.

For the Education Safari platform, various testing frameworks and methods were used to ensure both front-end and backend functionality. On the front end, Jest and React Testing Library were utilized to write unit tests for individual React components. These tests ensured that each component, such as the search bar, navigation menu, and resource listings, behaved as expected under different conditions. For instance, tests were written to verify that search results displayed correctly when a user enters a query or that the correct resource categories are displayed based on the selected filters.

On the backend, Mocha and Chai were used to perform unit and integration testing. These tests verified that the API endpoints built with Node.js returned the correct data and handled user requests properly. For example, tests were written to check that the API returned accurate search results based on the user's input and that the user authentication system worked as intended using JWT tokens.

4.8.4.1 Unit Testing

Unit testing involves testing the smallest parts of the system—individual components or units. In Education Safari, unit tests were performed on individual React components such as the Search Bar, Resource Card, and Navbar. These components were tested in isolation to ensure they behaved correctly in different scenarios. For instance, the Search Bar component was tested to verify that it correctly updates the state when a user enters a search query and triggers the search functionality.

Mock functions were used to simulate interactions with external dependencies, such as API calls. In this way, tests focused solely on the behavior of individual components without relying on the actual backend. For example, when testing the Resource Card component, the external data source was mocked, allowing us to focus on the card's rendering and interactivity. A unit test for the Search Bar component might check:

- That the search input updates the state correctly.
- That the search function is triggered upon pressing the search button.

4.8.4.2 Integration Testing

Integration testing ensures that individual units, when combined, work correctly together. In Education Safari, integration tests were written to test how different components interact, particularly the connection between the frontend and backend API. For instance, the Search Bar and Resource List components were tested together to verify that after a search query is entered, the Resource List is correctly updated with the fetched data. An example integration test was performed to:

- Simulate a user entering a search query in the Search Bar.
- Trigger an API call to fetch search results.
- Verify that the Resource List component updates with the correct resources based on the user's search query.

In integration testing, the API endpoints were not mocked, as the goal was to test the flow of data between the frontend components and the backend API.

4.8.4.3 System Testing

System testing involves testing the entire system as a whole to ensure it meets the functional requirements. For Education Safari, this was done through manual testing as well as end-to-end automated testing using Cypress. This type of testing ensured that all components—from searching resources to viewing resource details—worked together seamlessly from the user's perspective. During system testing, scenarios like these were tested:

- A user searches for a resource, views the results, and navigates to the resource details page.

A user navigates between different pages (e.g., home, search results, and resource details) and performs interactions such as saving a resource or filtering search results.

Table 2: Test Case_01

Test case ID	Test case description	Pre-condition	Test Steps	Test Data
TC_01	Check response when valid data are enter		1. Navigate to Contact Us page.	
		Valid Name.	2. Fill the Enter Name field.	Laxmi Regmi
		Valid email address.	3. Fill the enter email address field.	regmilaxmi325@gamil.com
		Valid contact number.	4. Fill the enter Contact Number field.	9824543632
		Valid messages.	5. Fill the message.	Testing Validation
Expected Result		Actual Result	Status	
Contact us page can be seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	pass	

Table 3: Test Case_02

Test case ID	Test case description	Pre-condition	Test Steps	Test Data
TC_02	Check response when invalid email address and contact number is entered.		1. Navigate to Contact Us page.	
		Valid Name.	2. Fill the Enter Name field.	Laxmi Regmi
		Invalid email address.	3. Fill the enter email address field.	egmilaxmi@gmail.com
		Invalid contact number.	4. Fill the enter Contact Number field.	982454
		Valid messages.	5. Fill the message.	Testing Validation
Expected Result		Actual Result	Status	
Contact us page can be seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	Pass	
Credential can be entered and seen.		As expected.	pass	
Failed message should be displayed.		A validation message is displayed saying enter valid email address and contact number.	Failed.	

4.8.5 Maintenance

Maintenance is a crucial aspect of any software project, including Education Safari. Even after the development phase, continuous maintenance is required to ensure the website remains functional, secure, and up-to-date. This involves tasks such as fixing bugs, adding new features, updating outdated content, and ensuring compatibility with new technologies and devices. As the platform evolves, the development team must be ready to incorporate new educational resources, improve the user interface, and optimize performance.

For instance, if the platform needs to support additional resource categories or change its search functionality, updates should be rolled out smoothly. Regular updates to the UI for improving user experience and managing user roles will also be essential. Additionally, system maintenance may involve training new users (students or administrators) on how to use the platform effectively, ensuring the system's long-term usability and adaptability to new user needs.

4.9 Activities Done

As part of my work on the Education Safari website, my primary responsibilities involved frontend development, focusing on creating a responsive and user-friendly interface. My tasks included implementing the UI components, developing the search functionality, and ensuring smooth navigation across the website. I worked closely with the project team, including project managers, designers, and backend developers, to align my work with the overall goals of the project.

The specific activities I carried out during the internship include:

- Study of existing educational platforms to understand common design patterns and identify areas of improvement for Education Safari.
- Wireframing and Prototyping: I created low-fidelity wireframes to map out the user flow and high-fidelity mockups using Figma, ensuring the platform's layout was intuitive and visually appealing.
- Developing UI components: I used React.js for creating modular and reusable components, such as the NavBar, Search Bar, and Resource Cards.
- Responsive Design Implementation: Tailwind CSS was employed to ensure that the website worked seamlessly across different devices, from desktops to smartphones.
- Collaborating with Backend Developers: I frequently communicated with backend developers to ensure that the frontend was integrated correctly with the API and that the search functionality worked as expected.
- Usability and Performance Testing: I conducted testing to ensure that the website was both user-friendly and performant, especially when handling large datasets of educational resources.

- **Feedback and Iteration:** Based on feedback from team members and users, I made necessary adjustments to the UI, ensuring it met user expectations and improved the overall experience.

4.9.1 Daily Meeting

As part of the agile development process for Education Safari, daily stand-up meetings were conducted to track progress and address any challenges. These meetings allowed team members to update each other on their current tasks, discuss any blockers they were facing, and plan for the upcoming tasks. I regularly attended these meetings to share my progress on the front-end development and coordinate with backend developers and designers to ensure smooth workflow integration.

4.9.2 Sprint Turnover

The development process for Education Safari followed an agile methodology, divided into 2-week sprints. At the end of each sprint, tasks were reviewed, and any incomplete work was carried over to the next sprint. This iterative process allowed the team to adapt to changes and incorporate feedback, ensuring the website was built incrementally and improved after each iteration. My contributions during each sprint focused on refining the UI components and ensuring they were ready for integration with the backend.

4.9.3 Retrospective

At the end of each sprint, retrospective meetings were held to evaluate the sprint's progress. The team discussed what went well, what could be improved, and how the workflow could be optimized for the next sprint. In these meetings, I shared insights on the frontend development challenges I faced, such as ensuring responsive design across multiple devices and improving page load performance and worked with the team to identify solutions for future sprints.

4.9.4 Live Demonstration

After each sprint, a live demonstration was conducted to showcase the current progress of the Education Safari website. I presented the newly developed UI components, demonstrated how the search functionality worked, and explained the responsive design implementation. These live demos were crucial for gathering feedback from stakeholders and ensuring that the website aligned with their expectations.

4.9.5 Documentation and Testing

Throughout the development process, I was responsible for documenting my work and ensuring that the front end was thoroughly tested. I maintained a work log to track the completion of tasks, including designing components, integrating them with the backend, and conducting usability testing. I also collaborated with the team to ensure that any bugs or issues identified during testing were addressed before the next sprint began.

4.10 Weekly log

Weekly logs were maintained to keep track of the work done during the week. The logs included roles and responsibilities assigned, activities performed, observations, and plans for the upcoming week. The performance appraisal by the team lead was also included in the log. The weekly log provided a high-level overview of the work done but was not an exhaustive list, as it may not contain all the work performed during the week. The maintenance of weekly logs helped in keeping track of progress and made it easier to review the work done during the week

The weekly logs served as a structured record-keeping system during your internship, capturing the key tasks and milestones achieved each week. Each log entry detailed the specific responsibilities assigned for the week, helping to clarify your role in the project. The activities performed section provided a breakdown of the actual work done, from coding and design tasks to more strategic efforts like setting up components or collaborating with team members.

Additionally, the observations captured significant insights or feedback during the week. This might include challenges faced, creative breakthroughs, or learning moments that helped improve the project. Having these observations helped track both technical progress and personal development.

The plan for the next week section was crucial in setting goals and ensuring continuity in the project. It acted as a forward-looking strategy, enabling you to focus on upcoming tasks and stay aligned with the overall project objectives.

Finally, the performance appraisal by the mentor or supervisor provided formal feedback on your performance, allowing you to understand areas where you excelled and areas for improvement. This system of weekly logs not only helped you stay organized but also ensured effective communication with your mentors and team, allowing for continuous improvement throughout the internship.

Overall, maintaining these logs facilitated the tracking of progress, ensured accountability, and made it easier to review contributions and assess performance in a structured way, offering a clear record of your journey and achievements during the project.

Chapter 5

SUMMARY AND CONCLUSION

5.1 Findings

Over the course of my internship working on the Education Safari website, I gained significant practical knowledge and experience that built on the theoretical foundations from my academic coursework. Working in a real-world environment introduced me to the complexities and intricacies of developing and maintaining an educational platform, while also teaching me how to collaborate within a professional team. Key findings from my internship include:

- **Bridging the gap between theory and practice:** The experience of working on a live project helped me apply my academic knowledge, specifically in web development, to real-world problems.
- **Adapting to a professional work environment:** I learned how to work within a team, manage deadlines, and deal with feedback from multiple stakeholders.
- **Understanding the importance of user experience (UX) in design:** By working on the frontend, I learned the significance of creating a seamless and intuitive user interface that enhances the user's experience.
- **Exposure to agile methodologies:** I gained hands-on experience with agile project management tools like JIRA and daily stand-up meetings, allowing me to better understand the iterative nature of software development.
- **Enhancing communication and collaboration skills:** Regular interaction with backend developers, project managers, and designers helped me develop my communication skills and understand the importance of coordination in achieving project goals.

5.2 Conclusion

My internship with Education Safari has been an invaluable learning experience, providing me with a real-world platform to apply my skills in frontend development using technologies like React.js, Tailwind CSS, and HTML. As I was responsible for building and refining the user interface of the platform, I gained significant insights into the practical aspects of web development. This opportunity not only allowed me to hone my technical skills but also improved my ability to work under deadlines, coordinate with team members, and adjust based on feedback.

The internship experience has given me a clear understanding of the professional standards expected in the industry. I now feel more prepared to transition from an academic setting to a full-fledged professional environment, confident in my ability to contribute to future projects with a more structured and collaborative approach.

5.3 Recommendation

As Education Safari continues to evolve, there are several areas where improvements and additional features can be implemented to enhance the user experience and the platform's capabilities:

- **Mobile Compatibility:** One key area for future improvement is optimizing the website for mobile and tablet devices. Although the current design is responsive, creating a dedicated mobile app would provide a more streamlined experience for users who access the platform on smaller screens.
- **Enhanced Search and Filtering Options:** Additional filters and more advanced search functionalities could be introduced to allow users to refine their searches based on more specific criteria, such as course difficulty level, subject matter, or resource type.
- **Personalized Recommendations:** Implementing a system that provides personalized resource recommendations based on the user's search history and preferences could improve user engagement.
- **User Notifications:** Adding a feature to send users notifications, such as new resources being added or reminders for saved bookmarks, would increase user retention and interaction with the platform.
- **More Robust Testing and User Feedback Integration:** While the current testing covers essential features, increasing the scope of usability testing with real users will help identify and solve potential issues earlier, making the platform even more user-friendly.

As for the internship itself, one potential improvement could involve more structured mentorship and regular check-ins to ensure that interns feel supported throughout the process. While virtual communication was effective, face-to-face interactions would have strengthened the sense of teamwork and helped address any challenges more efficiently.

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APPENDIX

HomePage



[Home](#) [About Us](#) [Services](#) [Study Abroad](#) [Blogs](#) [Gallery](#) [Contact Us](#) [CONSULT NOW](#)

Welcome to

Education Safari

Welcome to Education Safari, registered education consultancy with a global presence with main offices in Australia, and Nepal. Our primary objective is to offer expert counselling and guidance to students who aspire to pursue their education in Australia and abroad.

Our company specializes in providing expert counseling and guidance to students aspiring to pursue education and migration services in Australia. Additionally, we offer English language classes in Chitwan, Nepal, enhancing students' linguistic skills for successful international studies.

[VIEW MORE](#)



Our Services



Education Counselling

Our experienced team of counsellors is dedicated to understanding the unique needs and aspirations of each student. We provide personalized counselling sessions to help students explore suitable courses, institutions, and career pathways in Australia.



Admission Assistance

Navigating the complex admission processes can be daunting for international students. We assist in preparing and submitting applications to Australian universities and colleges, ensuring that all necessary documentation is complete and accurate.



Visa Guidance

Securing an Australian student visa is a crucial step in the journey. Our visa experts guide students through the visa application process, making it as smooth and stress-free as possible.



Pre-Departure Support

We understand that moving to a new country can be overwhelming. To ease the transition, we provide pre-departure briefings, helping students with essential information about life in Australia and preparing them for the new experience.

Expert Counselling and Guidance

Our primary objective is to offer expert counselling and guidance to students who aspire to pursue their education in Australia and abroad.

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Chitwan Office



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Narayanganj, Chitwan,
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Admission Assistance
Visa Guidance
Pre-Departure Support
English Language

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Telephone : 08 8472 8260
Mobile phone : 0424 640 356
Email : info@educationsafari.com

WORKING HOURS

MONDAY: 9 AM to 6 PM
TUESDAY: 9 AM to 6 PM
WEDNESDAY: 9 AM to 6 PM
THURSDAY: 9 AM to 6 PM
FRIDAY: 9 AM to 6 PM

Responsive View



Education Safari
Education | Migration



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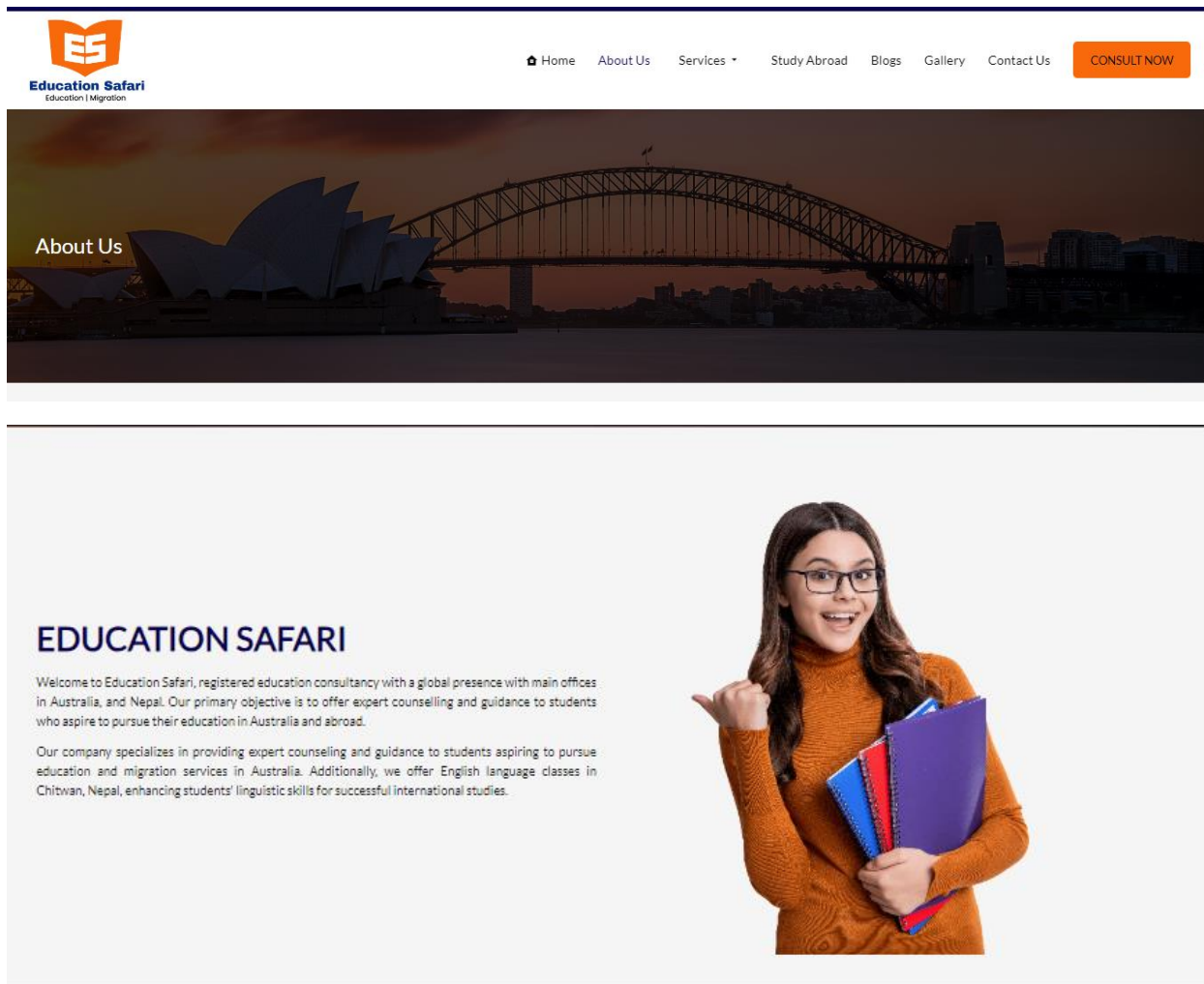
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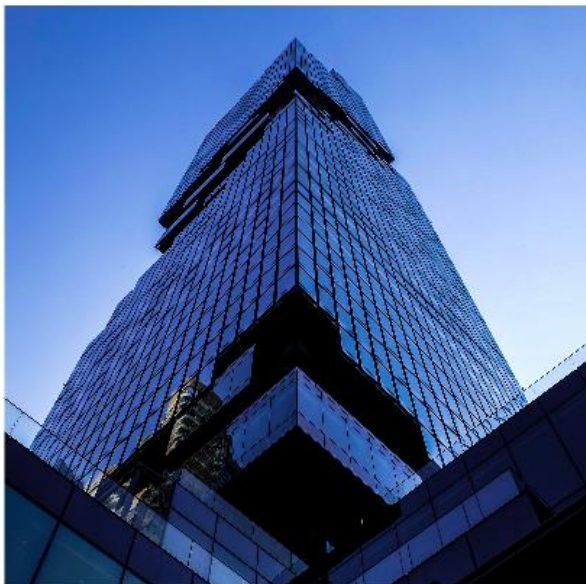
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About Us page





Why Choose Us?

Expertise

Our team consists of seasoned education consultants with in-depth knowledge of the Australian education system and immigration policies. We are well-equipped to address students' concerns and provide accurate information.

Personalized Approach

We believe in understanding each student's goals and preferences to tailor our services accordingly. Our personalized counselling ensures that students receive guidance that aligns with their aspirations.

Extensive Network

Over the years, we have established strong connections with reputable educational institutions in Australia. This network enables us to facilitate seamless communication between students and universities.

Commitment to Success

We measure our success by the achievements of our students. Your dreams are our priority, and we are dedicated to supporting you at every step of your educational journey.



Our vision

To be the preferred global education consultancy, empowering students to achieve their academic and career aspirations through our personalized guidance and expert services, particularly for Australia.

OUR MISSION

To provide unparalleled educational counselling, migration services, and language enhancement programs that bridge the gap between students' dreams and the reality of studying in Australia. We aim to be the guiding light for students seeking a transformative international education and migration experience.



Our Team

 Er. Sagar Shrestha Director MIT (Pindara University) QEAC #12075	 Shyam Dahal Operation Manager MAss (University of Western Sydney) MBS (TU) GDIP in MBS (Pindara University) Bachelor of Nursing (University of South Australia) QEAC #12091	 Khagendra Sapkota Finance Manager MPA, Registered Tax Agent Master of Commerce (Professional Accounting) University of Ballarat Australia QEAC #12103
 Surya Acharya Business Development Manager Bachelor of Nursing (UNISA) Master of MHN (University of Newcastle) QEAC #12072	 SK Bohara Marketing Manager Bachelor of Accounting (University of Ballarat) Bachelor of Nursing (Pindara University) QEAC #12092	 Yatharth Bharadwaj Registered Migration Agent MARA 5680071

Services page

Our Services

 Education Counselling Our experienced team of counsellors is dedicated to understanding the unique needs and aspirations of each student. To ensure successful outcomes, we provide a range of services across the entire education and immigration process.	 Admission Assistance Highlighting the various admission processes on the Australian education system, we assist students in choosing the right course and university, ensuring that all relevant documentation is completed accurately.	 Visa Guidance We provide Australian student visa guidance and the correct documentation to ensure successful visa applications. Our experienced team provides ongoing support throughout the entire process, from initial application to arrival in Australia.	 Pre-Departure Support We understand that moving to a new country can be challenging. To ease the transition, we provide pre-departure support, including airport transfers, accommodation, and assistance with settling into the new environment. Our team is always ready to help with any queries.
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Expert Counselling and Guidance

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Students' Feedback





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Pre-Departure Support

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English Language

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Admission Assistance

Visa Guidance

Pre-Departure Support

English Language

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Enter your full name

Email*

Enter email

Phone*

Enter phone

Service*

Education Counselling

What date and time work best for you?*

2024-09-17

Tuesday, September 17

11:45 AM12:45 PM



Admission Assistance

Navigating the complex admission processes can be daunting for international students. We assist in preparing and submitting applications to Australian universities and colleges, ensuring that all necessary documentation is complete and accurate.

Fill the form below

Full Name *

Email *

Phone *

Service *
Admission Assistance

What date and time work best for you? *
2024-09-17

Tuesday, September 17
11:45 AM 12:45 PM
01:45 PM 02:45 PM
03:45 PM 04:45 PM
05:45 PM

Message *
Your message here.

Submit

Study Abroad Page



Study Abroad



Study in Australia

Australia is very well known for being one of the top choice for students who want to pursue their studies in a foreign country.



Study in Canada

The country of maple syrup, ice hockey and northern lights is famous for not only these wonderful things but also for the welcoming and polite nature of people.



Study in UK

United Kingdom is famous not only for the biggest Empire in the history they had and for having towns with huge castles.

The country of maple syrup, ice hockey and northern lights is famous for not only these wonderful things but also for the welcoming and polite nature of people. Canada is home to many immigrants from all over the world which makes the country richer in cultural and religious aspect. The country welcomes people of all races, religion and ethnic background which makes the country great for any international student wanting to study in Canada.

Canada is second largest country in the world by total area and is located in the northern part of North America. It comprises of three provinces and ten territories. Canada has two official language: English and French, and its official currency is Canadian Dollar. The country has constantly ranked high in terms of quality education and standard of living. Canada also ranks 11th for the highest GDP in the world.

For international students who are interested in studying in Canada but have confusion in any of the process can directly contact our team of experts, they will help find answers to your questions.

Benefits of Studying in Canada

Canada is attractive to the people not only for the great outdoors and eye-catching attractions, but for the great quality of life they get to enjoy. There are many advantages for the international students planning to study in Canada, it is a multicultural country that gives importance to individual freedom. If a student is concerned about their safety while studying abroad then Canada is a great choice for them. The country being the second largest in the world has relatively lower crime rates.

Canada provides great quality of education and has top universities in the world. Canadian universities are world renowned for innovation and research. Getting higher education from a Canadian University will help your professional as well as personal life, you will get to meet students from all over the world and gain experience in the process that will benefit you in the long run.

Degree from Canada will help you get recognized in the competitive market. Canada also has relatively lower tuition fees compared to other developed countries that provide quality education. And while studying in Canada international students can work up to 20 hours per week, this helps students maintain their lifestyle while studying. Canada also offers post-study work permit to the students where the students can apply for work permit after they graduate up to 3 years.

Education System

Canada is known for being the most educated country in the world, having the literacy rate of 99%. Education System can differ in each province but there is compulsory education in all of the provinces. The Education System is mainly divided into three levels.

Primary Education: This level runs for pupils from age 6 to 7 and goes till they are 12 to 14 years old that is Grade 8.

Secondary Education: This level runs from grade 9 to grade 12, which is also known as high school.

Post Secondary Education: For this level of education students can choose the university and course of their liking. Canada has internationally recognized universities that offers various courses where students can choose the course that they like, here the length of the course depends on the course they select.

Cost of Studying in Canada as well as Scholarship opportunities

Having multicultural cities and great education system, Canada welcomes more than 350,000 international students each year who aims to fulfill their dreams studying in one of the most known country for quality education. Being great in terms of education, it is well known study destination for students and Canada is comparatively affordable than other developed countries that provides quality education. So, it is a great choice for students who want to study within a budget. Generally you will need about C\$20,000 and C\$30,000 annually to cover your tuition fees.

Cost of studying varies on what you decide to choose and length of the course. For undergraduate program in Education field you will need around \$20,875 whereas for Dentistry it can go up to \$56,621. Universities in Canada offers various scholarships to international students, and if you want to apply for it and learn more about it you can book an appointment with us so we can help you find the right scholarship for you.

Living and Working in Canada (Employment Prospects)

Quality of living and employment opportunities in Canada attracts immigrants from all around the world. You will be able to meet and interact with individuals from every corner of the world while you study there. Canada welcomes people from different races, ethnic groups and is a hub for international students. Living cost in Canada greatly varies on the fact where you plan on staying. For example, the cost of one-bedroom apartment in Toronto can cost 1,400 CAD per month more than in Québec.

The more you stay near popular areas like Toronto, Vancouver the more there is chance of finding accommodations that are expensive. As for job opportunities you will be able to find one depending on your skills and degree that you've acquired. Even

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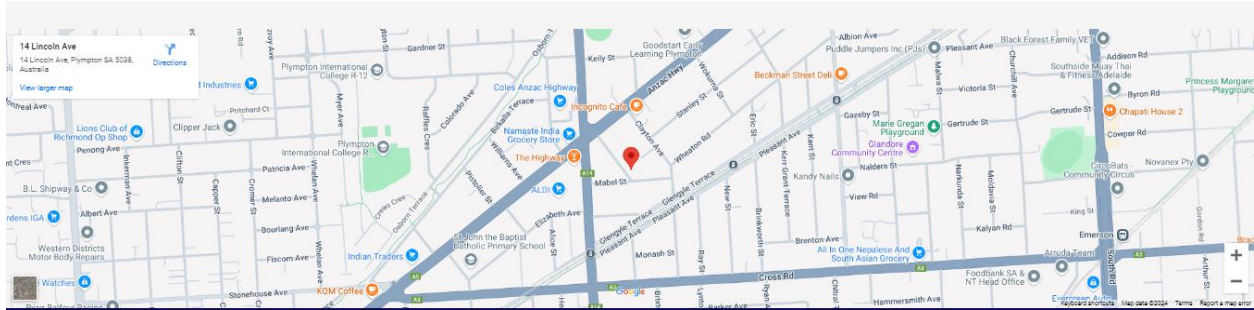
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Please Fill The Information Below

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Phone *

Service *

-- Select Service --

What date and time work best for you? *

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11:45 AM

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